How to find love in the Internet? Applying Web mining to affect recognition from textual input.

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Abstract. A method for affect analysis of textual input in Japanese is proposed. In the process of analysis, first the emotiveness of an utterance is determined and then the specific types of emotional states conveyed in an utterance are extracted. We support the system with a Web mining technique to improve the performance of emotional state types extraction. The system constructed on the basis of the proposed method reached human level in determining emotiveness of utterances and 63% of human level in extracting specific types of emotions. The supporting Web mining technique improved the performance of the emotional states types extraction to 85% of human performance.

Keywords: Affect analysis, emotiveness, Web mining.

1 Introduction

Technological development has led to the creation of a new dimension of communication, where a machine is one part, man-machine communication (MMC) [1]. A rush in development of talking robots and conversational systems [2] indicates that the functional implementation in our lives of agents like intelligent car navigation systems [3] or talking furniture [4] has already become a current process, and a need for humanized interfaces in MMC grows rapidly.

One of the most important cognitive human behaviors present in everyday communication is expressing and understanding emotions. People make many decisions under the influence of their own emotions or the emotional states of others [6, 7]. Therefore one of the current issues in fields like agent development or communication and computer science is to produce methods for effectively recognizing and analyzing users' emotional states.

Recent introductions in our lives of technologies such as 3G, ubiquitous networks or wireless LAN made the Internet, a social phenomenon only a few years ago [8], an indispensable "everyday article". There was a rapid increase of fields using Internet resources as an object of research, like information or opinion retrieval [9]. The Web is becoming a determinant of human commonsense [10]. Within the last few years there were several trials to retrieve information concerning human emotions and attitudes [11] from the Internet. However, there have not been many attempts to use Web mining for affect analysis.

2 Affect Analysis

Affect analysis is a sub-field of information extraction. Its goal is to estimate human emotional states in communication. Popular methods for analyzing affect include analyzing emotions from facial expressions or voice [5]. However, since emotions are strongly context dependent, most of the semantic content of expressing emotions is ignored in such research.

Therefore, we decided to analyze the affect of textual representations of utterances. Furthermore, our method of affect analysis uses the advantages of Internet resources widely available today.

Three-step Approach to Affect Analysis

Many attempts to analyze the emotive aspects of text end in a failure. Read [12] helplessly strived against the lack of an appropriate emotive lexicon and the unsystemized classification of emotion types in the end, achieving an accuracy of 33%[1]. Alm et al. [13] achieved an accuracy of 69%. They too wrestled head on with the lack of emotive databases, but also with an inappropriate evaluation corpus consisting of children's stories full of ambiguities arising from mixing styles and means of expression (e.g. dialogues mixed with narrative style and descriptions). Finally, Wu et al. [14] achieved 72% of general average accuracy, but ran into a problem with ambiguity of emotional rules. Another problem is the evaluation of such systems. In much research the means of evaluation are unstandardized, performed with a small number of evaluators [35] with crucial parts of the evaluation performed by the researchers themselves [14].

Therefore we made the following assumptions to specify what we mean by affect analysis. The research will be conducted on the basis of textual utterances. Compared to using the large and quickly rising number of textual corpora, it is time consuming and uneconomical to gather a relevant corpus for the analysis of vocal and visual dimensions. Moreover, neglecting semantic meaning in such research leads to ambiguities in emotional state determination since some realizations of non-verbal communication, like high pitch or arousal can convey different emotional states. Restricting ourselves to a textual corpus thus gives a green light to concentrate on emotiveness realized in the semantic surface of the text. The analysis will be conducted on dialogue-like utterances. This approach relates to the future goal, which is to implement the system into a conversational system like Higuchi's [2] to make the utterance understanding and generation more natural. The language we will work with is Japanese. The analysis of affect is performed in three steps: 1. Determining the emotiveness of an utterance, or finding whether an utterance is emotive or not; 2. Finding how strong the emotions conveyed in the utterance are, or setting the emotive value; 3. If the sentence is emotive, determining what types of emotions are conveyed in the utterance.

[1] All results approximated to the nearest whole percent.

3 Emotiveness - a Linguistic Approach to Emotions

In an everyday conversation there are different linguistic and paralinguistic means used to inform other interlocutors of emotional states. The elements of speech used to convey emotive meaning in sentence make up the feature of language called emotiveness [15] and the function of language describing them is called the emotive function of language [16, 17].

The emotive function of language is realized lexically in Japanese through such parts of speech as the following: exclamations [18, 19], hypocoristics (endearments) [20], vulgar language [21] or mimetic expressions (*gitaigo*) [22]. A key role in expressing emotions is also played by the lexicon of words describing states of emotions [23]. The analysis of elements of language such as intonation or tone of voice as well as nonverbal elements, like gestures or face expressions, in applications such as ours where the communication channel is limited to transmission of lexical symbols, has to be focused on its textual and lexical manifestations, like exclamation marks or ellipsis.

4 Definition and Classification of Emotions

In the field of affect analysis there is a tendency to forge definitions and classifications of emotions for the needs of a particular research project. This might arouse doubts as to whether the classification was not done to match the results. To avoid this problem and to contribute to standardization of criteria, we decided to use definitions and classifications of emotions based on the most thorough research in this matter currently available for the Japanese language [23] rather than create our own.

Definition of Emotions

We use the general definition of emotions proposed by Nakamura, who defines them as every temporary state of mind, feeling or emotional state evoked by experiencing different sensations [23]. This definition is complemented by Beijer's definition of emotive utterances, which he describes as "every utterance in which the speaker in question is emotionally involved, and in which this involvement is linguistically expressed by means of intonation or by the use of performative expressions." [18]

Classification of Emotions

Nakamura [23], after a thorough study on emotions in Japanese, proposed a classification of emotions into 10 types - said to be the most appropriate for the Japanese language. That is: *ki / yorokobi* (joy, delight), *do / ikari* (anger), *ai / aware* (sorrow, sadness), *fu / kowagari* (fear), *chi / haji* (shame, shyness, bashfulness), *kou / suki* (liking, fondness), *iya / iyodomi* (dislike, detestation), *kou / takaburi* (excitement), *an / yasuki* (relief) and *kyou / odoroki* (surprise, amazement).

5 Emotive Elements/Emotive Expressions Analysis System

Based on the linguistic approach and assumptions described above Ptaszynski et al. [24] constructed ML-Ask (Emotive Elements / Emotive Expressions Analysis System) for analyzing the emotive contents of utterances. The system uses a three-step procedure: 1) Analyzing the general emotiveness of an utterance; 2) Calculating the emotive value representing the strength of the conveyed emotions; 3) Recognizing the particular emotion types. The system is based on Ptaszynski's [25] idea of two-part analysis of realizations of emotions in language into:

1. *Emotive elements*. Indicating that emotions have been conveyed, but not detailing what specific emotions there are. The same emotive element can express different emotions depending on context. This group is linguistically realized by such subgroups as interjections, exclamations, mimetics, or vulgar language, described earlier in section 3. Examples are: *sugee* (great!), *wakuwaku* (heart pounding), *-yagaru* ("fu**ing do sth" – a vulgarisation of a verb);
2. *Emotive expressions*. Parts of speech, that in emotive sentences describe emotional states. This group is realized by various parts of speech, like nouns, verbs, adjectives, etc. Examples are: *aijou* (love), *kanashimu* (feel sad), *ureshii* (happy), respectively.

However, for some of the emotive elements the ambiguity of the emotional affiliation is restricted to only some emotion types.

The emotive element database was built using data from different research [22, 26, 27, 28, 29] and divided into interjections, mimetics, endearments, vulgarities, and representations of non-verbal emotive elements, such as exclamation marks or ellipsis. An algorithm detecting emoticons was also added, as they are symbols commonly used in everyday text-based communication tools. The database of emotive expressions contains Nakamura's collection [23].

5.1 Analysis Process

For a textual input provided by the user, three features are calculated in order: general emotiveness, emotive value, and specific type of emotion. Firstly, the system searches for emotive elements in the utterance to determine the general emotiveness of the utterance (emotive / non-emotive). Secondly, if any emotive elements are found, the system calculates the emotive value as the sum of all emotive elements found (but with a maximum value of 5). Thirdly, in emotive utterances the system searches for emotive expressions to determine the specific type of emotion (or emotions) conveyed in the utterance. An example of analysis performed by ML-Ask is showed in Table 1.

This basic extraction procedure (see Fig. 1) is also extended with a Web mining technique described below in detail.

Table 1. An example of analysis performed by ML-Ask

Utterance	*Kono hon saa, sugee kowakatta yo. Maji kowasugi.* (That book, ya know, 'twas a killer. It was just too scary.)
Emotive elements	*saa, sugee, -yo, maji, -sugi* (**Emotive value** = 5)
emotive expressions	*kowai*

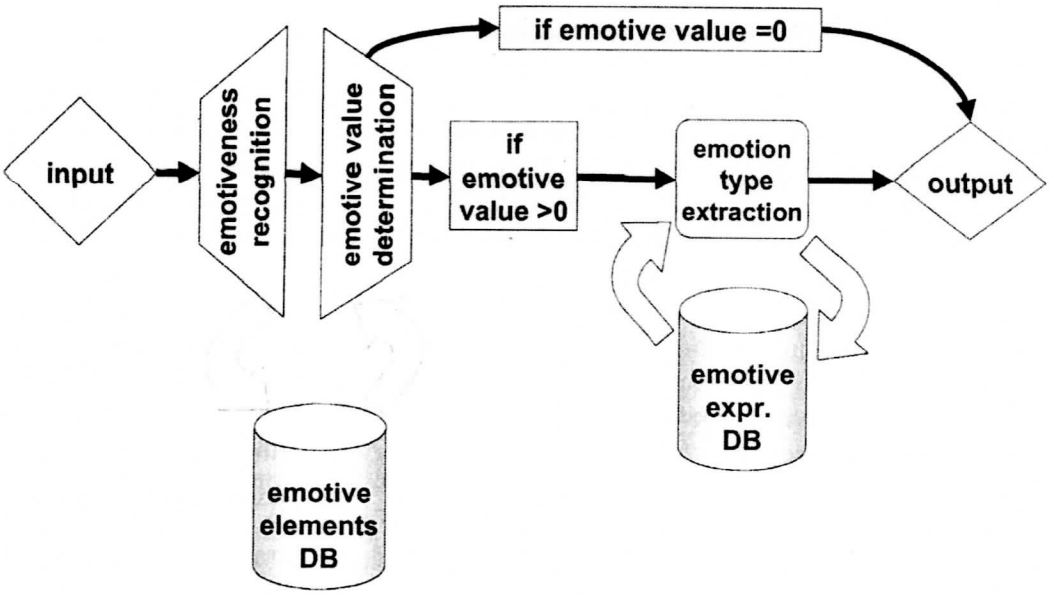


Fig. 1. Flow chart of the basic procedure for emotion type extraction in the ML-Ask system.

6 Web Mining Technique

To improve the extraction of the specific emotion types we apply Shi's Web mining technique for extracting emotive content from the Web [30] based on causality. However, since Shi's technique was effective only when a particular phrase to check in the Internet was extracted manually, we enhanced it with an automatic phrase extraction method based on dividing the sentence into separate n-gram phrases.

6.1 Phrase Extraction From an Utterance

An utterance is first processed by MeCab, a tool for part-of-speech analysis of Japanese [31]. Every element separated by MeCab is treated as a unigram. All the unigrams are grouped into larger n-gram groups preserving their word order in the utterance. The groups are arranged from the longest n-gram (the whole sentence) down to all groups of trigrams. N-grams ending with particles are excluded, since they gave too many ambiguous results in our pre-test phase. An example result of this sentence separating procedure is shown in Table 2.

Table 2. Example of n-gram separation from an utterance.

Original utterance	*Aa, pasokon ga kowarete shimatta*...(Darn, the PC has broken...)	
longest n-gram (here: hexagram)	*Aa pasokon ga kowareru te shimau* [interjection] [noun] [particle] [verb] [verb connector] [perfect form]	
pentagram	*pasokon ga koware te shimau*	
tetragram	*Aa, pasokon ga kowareru*	
trigrams	*pasokon ga kowareru*	*koware te shimau*

6.2 Emotion Type Extraction Using the Web

In the web mining the extracted phrase is used as a query to the Google search engine. Moreover, the n-grams ending with a verb or an adjective are grammatically adjusted for semantically deeper Web mining. They are modified in line with Yamashita's research on causality morphemes conveying emotive meaning [32] independently confirmed experimentally by Shi [30], as in Table 3. This morpheme modification provides additional query phrases. All the phrases, the extracted ones as well as the modified versions of the ones ending with a verb or an adjective, are queried in Google. A maximum number of 500 snippets for each queried phrase is extracted from the Web and cross-referenced with the database of emotional expressions described in section 5 (see Fig. 2). The emotive expressions extracted from the snippets are summed up, and the results for every emotion type are listed in descending order. This way a list of emotions commonsensically associating with the queried sentence is obtained (an example is shown in Table 4). In the evaluation experiment we compared two variants of the method: 1) extracting and keeping all emotions and 2) taking into account only the emotions that achieved the three highest results, treating the rest as a noise (e.g. in the Table 4. the four upper types are taken into consideration, since the two first have equal results).

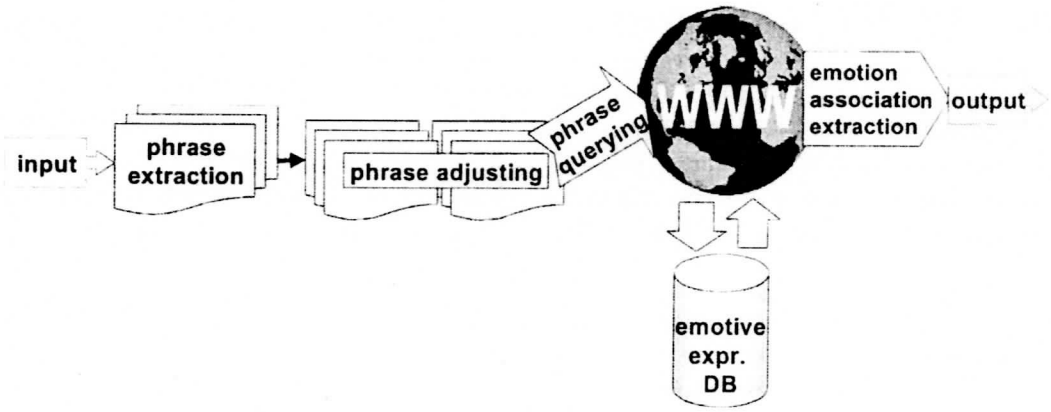


Fig. 2. Flow chart of the Web mining procedure.

Table 3. Examples of n-gram modifications for Web mining.

	Original n-gram	*pasokon ga koware te shimau*
n-gram phrase adjusting (morpheme modification)	*/ -te /*	*pasokon ga koware te shima- tte*
	/ -to /	*pasokon ga koware te shimau to*
	/ -node /	*pasokon ga koware te shimau node*
	/ -kara /	*pasokon ga koware te shimau kara*
	...	...

Table 4. Example of emotion types extraction from the Web

Sentence: *Aa, pasokon ga kowarete shimatta...*(Darn, the PC has broken...)		
Extracted emotion type	**Type extracted / all extracted types**	**Ratio**
[fear]	28/133	0.210526315789474
[sorrow, sadness]	26/133	0.195488721804511
[dislike, detestation]	16/133	0.120300751879699
[liking, fondness]	14/133	0.105263157894737
[relief]	12/133	0.090225563909774
[excitement]	11/133	0.082706766917293
[joy, delight]	10/133	0.075187969924812
[surprise, amazement]	9/133	0.067669172932330
[anger]	5/133	0.037593984962406
[shame, shyness, bashfulness]	2/133	0.015037593984962

7 Evaluation Experiment

To determine the system's accuracy we performed an evaluation experiment. In the experiment we evaluated the ML-Ask system's performance for the whole three step procedure described in section 5 and compared the results for emotion type extraction for three different extraction procedures: 1) ML-Ask basic procedure; 2) ML-Ask with Shi's technique combined; 3) ML-Ask with Shi's technique instead of the usual emotion extraction method (see Fig.3). For each extraction procedure we also checked the two variants mentioned in section 6.2 – taking into consideration all of the extracted emotions or leaving only the ones with the 3 highest results discarding the rest as a noise.

As evaluation method we use the Double Standpoint Evaluation Method (DSEM) for affect analysis systems [33] proposed by Ptaszynski et al. that has been shown to be objective in comparison to the usual methods. DSEM is a method where the evaluated system is watched from two different standpoints: recognitive (the first person evaluation) and commonsensical (the third person evaluation). As a method aiming to be objective, it assumes that neither do people themselves understand their emotional states with 100% reliability, nor do other people perceive emotional states of their interlocutors with a perfect accuracy. In the method, the system is first evaluated on whether it can appropriately recognize the emotional states of users. After that, another evaluation is performed by an objective group of third party evaluators to check how much the system's procedures agree with human commonsense about other person's emotions. For both sets of results, 1) the higher the results of both first- and third-person evaluation are, the better, and 2) the more balanced the both results are, the better.

The evaluation is based on a corpus of natural utterances gathered through an anonymous survey. In the survey 30 people from different ages and social groups participated. Each of them was to imagine or remember a conversation (or conversations) with any person (or persons) they know and write three sentences from that conversation: one free, one emotive, and one non-emotive. After that the authors tagged the utterances written by themselves in the same way as the system procedure - first whether an utterance is emotive. If so, the authors set the emotive value (0-5) and describe the specific emotion types conveyed in the emotive utterances.

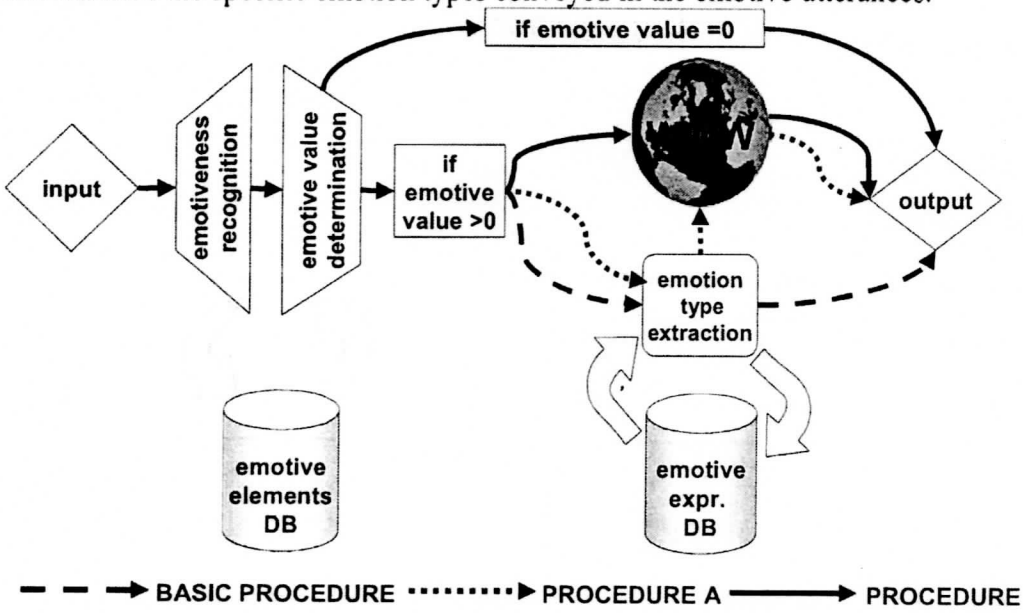


Fig. 3. Flow chart of the ML-Ask system with 3 procedures of emotion types extraction.

We used ML-Ask to analyze this corpus of 90 sentences, and compared the results of the system with the results provided by the authors of the utterances. Although the ideal would be a perfect agreement, it is highly difficult to achieve such a score, since attitudes towards emotive features of speech differs greatly between people. Therefore, we also added an emotive tagging of the corpus by third party human evaluators to determine a general human level in recognizing emotions from text in ordinary people. The third party human evaluators (on average 10 people per sentence) tagged the corpus emotively in the same way as the system does and for each of them we counted the accuracy in recognition by comparing the results to the results provided by the authors of the utterances. The level of results achieved by the third party evaluators was considered to be the human level of accuracy in recognizing emotions from text.

This provides us the evaluation from the first (recognitive) perspective. To broaden the evaluation, the second (commonsense) perspective is also applied. By having the same utterances tagged by third party human evaluators (10 people on average) and calculating the system's agreement with them, we can check the commonsense level of the evaluated method. In our assumption, the more similar the system was to the

average impressions of the people, the closer it was to the level of human commonsense.

We calculate the agreement coefficient U between one evaluator (or author of an utterance) Eva_A and another Eva_B (or system) as a simple ratio of the number of identically tagged utterances t_{sim} to all of the utterances tagged by the human evaluator t_i, as in equation 1.

$$U_{Eva_B}^{Eva_A} = \frac{t_{sim}}{t_i} \tag{1}$$

8 Results and Discussion

Emotive / Non-emotive. The system's accuracy in determining the emotiveness is calculated as the harmonic mean of the accuracies on emotive and non-emotive utterances.

The total accuracy of ML-Ask in this matter evaluated on 90 items was 0.83. This result is very promising, since the same value calculated for the third party human evaluators gave a wide range of results from 0.4 to 0.86. The result of ML-Ask was placed close to the top ranking like this, and therefore we can say that the system recognizes emotiveness on a very high human level.

In the commonsense evaluation the method achieved 86% of the human level of commonsense (human interagreement). The result, while not reaching the full human level of commonsense, is still very promising.

As an interesting fact, we should add that for sentences with a perfect match, where both authors and all evaluators agreed on the emotiveness (18 sentences), the system's result was 100%.

Emotive Value. The many constantly changing situational features of a conversation as well as personal experiences of interlocutors make the emotive value of an utterance difficult to decide strictly on a scale 0 to 5.

Since the harmonic mean is thus inappropriate for this stage of evaluation, we apply the agreement coefficient. We also use a condition of almost-perfect match, when the emotive value differs between the author of an utterance and the system (or between two evaluators in general) by only ±1 emotive point.

For 30 appropriately tagged items for this stage of the evaluation the system achieved an U of 0.53. The accuracy in judging the emotive value by third party evaluators was U=0.84. The system's result thus reaches 63% of the human level accuracy (0.53/0.84). Thus, it is desirable both to find a more precise way of setting the emotive value in the future that is better than the quantitative one used this time, as well as to increase the number of third party evaluators to increase the reliability of the judgments.

In the commonsense standpoint evaluation, the system achieved 82% of the human level of commonsense in judging the emotive value. This result is close to, and therefore confirms the result of the general emotiveness evaluation (86%). Although it is desirable to improve the method for judging the emotive value to reach the human

commonsense level, we consider the result promising, since the emotional intensity setting is highly subjective and ambiguous among people. It is also likely that improving the accuracy of determining the emotiveness of a sentence by eliminating the ambiguities in emotional affiliations of emotive elements will positively influence the emotive value setting accuracy.

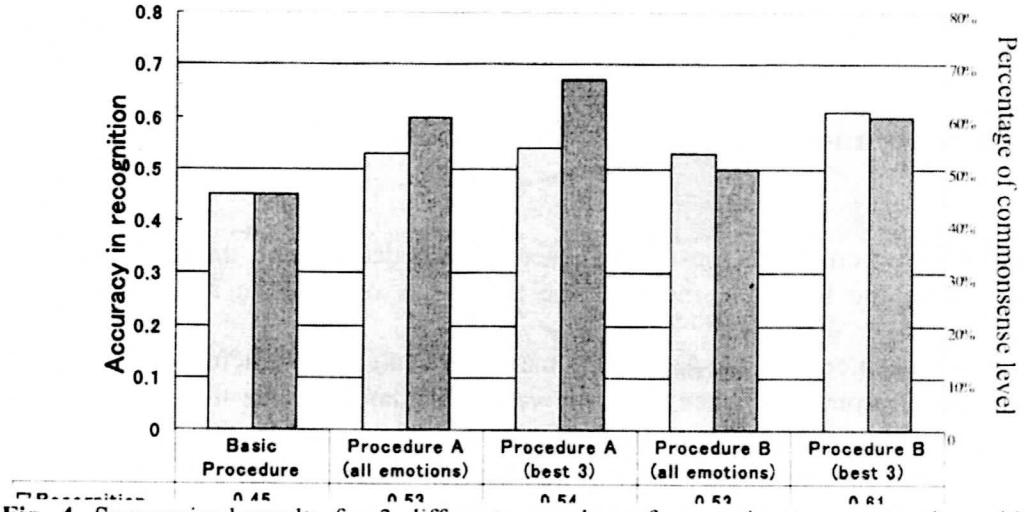


Fig. 4. Summarized results for 3 different procedures for emotion type extraction with 2 variant procedures using Web mining techniques.

Specified Emotions. The system can potentially extract many emotion types (up to ten) for one utterance. However some of them can be extracted wrongly, and there is a possibility that there would still be some emotion types not extracted. Therefore we decided to calculate the system's results as the balanced F score with the emotive tags added by the authors of the utterances as a gold standard. For the commonsense evaluation we use all emotion types tagged by the third party evaluators as a set of emotion types commonsensially associating with a particular utterance. Correctly specified emotion type is understood as recognizing any and at least one emotion type from all of the set, including non-emotive (the results are summarized in Fig. 4.).

Basic procedure. The system's accuracy in estimating the specific types of emotions reached balanced F-score of 0.45. As for human evaluators, the average accuracy was 0.72. Therefore the system's accuracy reached approximately 63% (0.45/0.72) of the human level. In future research it is desirable to both improve the algorithm for determining the specific emotion types and to perform the evaluation on a larger number of tagged utterances. In the commonsense evaluation the basic method for emotion extraction reached 45% of the human level.

Procedure A. This method for emotion type extraction used Web mining technique as a support for ML-Ask in all cases where the system could determine that an utterance was emotive but after performing the basic procedure was not able to determine the

emotion type. ML-Ask supported with this method achieved a balanced F-score of 0.54 and 0.53 for the two variants "3 best results" and "all emotions" respectively. This gives approximately 75% (0.54/0.72) and 74% (0.53/0.72) of the human level in determining the emotion types. The commonsense level of this method was evaluated to 67% and 60% respectively.

Procedure B. In this method for emotion type extraction, the Web mining technique is used instead of the system's lexical extraction (basic procedure). ML-Ask with this method achieved balanced F-score of 0.61 and 0.53 respectively of for the two variants. Compared to the human level in determining the types of emotions this amounts to 85% (0.61/0.72) and 74% (0.53/0.72) respectively. The commonsense level of this method was evaluated to 60% and 50% respectively.

9 Conclusions and Future Work

In this paper we presented a method for affect analysis of utterances, supported by a Web mining technique. In every case the Web mining improved the accuracy of the system in extracting the specific types of emotions, although there were some differences in commonsense levels found for the two extraction procedures we checked in this paper. There is a need for further experiments on a larger evaluation material to determine which is the most accurate extraction procedure. Experiments on different variants of the method showed that it is more effective to keep only the emotions that achieved the three best results and discard the rest as a noise, rather than keeping all of the extracted emotions. Furthermore, since the system's procedures are used sequentially, improving accuracy in earlier stages is very likely to improve the emotion type extraction. The tools for morphological analysis used in the system are also not perfect, which might decrease the system's real accuracy. However improving the extraction of n-gram phrases from the sentence to analyze should eliminate these difficulties. In the near future we also plan to apply Russell's two-dimensional model of emotions [34] to reduce the ambiguities in the database of emotive elements, which should also enhance the emotion type extraction. Also, the processing time in the procedures using Web mining is rather long (several minutes or more), which might be caused by potential problems with network connections or imperfections in search engine indexing. However, this can be mitigated by using a different search engine, or applying a Web page indexing method created especially for the needs of this research, for example using HyperEstraier [36], which we plan to do in the near future.

Acknowledgments. This work was partially supported by a Research Grant from the Nissan Science Foundation.

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